

Biçimsel Diller ve Soyut makineler

-Regüler Dil

Kuralları:

- ϵ, λ boşluk katarı regülerdir
- ϵ Alfabe de bulunan her harf regülerdir.
- A ve B reguler ve; $A \cup B, AB, A^*$ regülerdir.

-Katar uzunluğu

- $|W| = 5 \rightarrow W = abcde$
- Boşluk karakter katarının uzunluğu sayının $|\lambda| = 0$ ~~***~~

concatenation

- $W = abba$; $V = bbbaaa$ • $WV = abbabbbbaaa$

Direct or'da buluşturan

-Reverse

- $W = abcde$ • $W^R = edcba$

-Bitiştirme

- $|UV| = |U| + |V|$ • katar bitiştirme oluşan katarın uzunluğu a'yı a'yı uzunluklarının toplamıdır.

-eşit olması

- Katarının eşit olması için bir bire aynı olması gerek

-üst alma

- $(abc)^2 = abcabc$ • $W^0 = \lambda$ • $(abba)^0 = \lambda$

-Kleene yıldızı

- alfabe üzerinde oluşturabilir tüm katarların olduğu küme

Kleene yıldızını oluşturur

- mutlak boşluk karakter olmalı

Regüler diller ve regüler ifadeler (CDEA, NFA, $\wedge$, $\rightarrow$, $\dots$)

- Pozitif Kapanma

- (ϵ^+) Kleene yıldızından tek farkı boşluk karakteri katarı çıkarır.

- Dilin reverse

- $\{ab, aab, baba\} \rightarrow \{ba, baq, abab\}$

Dil Bitirme

- $L_1 = \{a, ab, ba\}$ • $L_2 = \{b, aa\}$
- $L_1 L_2 = \{ab, aaa, abb, \dots\}$

- Dil üstüne alma

- $\{a, b\}^3 = \{aaa, aab, aba, \dots\}$

Dikkat $L = \{a^n b^n, n \geq 0\}$, $L^2 = \{a^n b^n a^m b^m, m, n \geq 0\}$

- Yıldız Kapanma

- $L^* = L^0 \cup L^1 \cup L^2 \cup \dots$ • $\{a, b\}^* = \{\epsilon, a, b, ab, \dots\}$

- Pozitif Kapanma

- Sadece boşluk karakter katarı yok.

- $\{\epsilon\}$ Boş küme regüler bir dildir.

- $\wedge$ regüler bir dildir.

- Σ her semgisi regüler bir dildir.

- A, B regüler bir dil ise $A \cup B, A^*, A.B$ regüler bir dillerdir.

ÖR: $\{0\}$ ve $\{1\}$ regüler ise $\{0, 1\}$ regüler bir dildir.

- Yalnızca bir string (katar) içeren diller regüler dillerdir.

ÖR: $\{0, 1\}$ ve $\{00\}$ regüler ise $\{0, 1, 00\}$ dilde regüler bir dildir.

- tanımladığımız dilin eleman sayısı sonlu ise bu dil regülerdir.

Not: $\{0, 1\} \rightarrow \{0, 1\}^n$ regüler değildir.

Regüler diller ve Regüler ifadeler

operatorlerin öncelikleri:

Kleen yıldızı $>$ $()$ $>$ Bitleştirme $>$ Birleştirme

$(\epsilon 0^* \cup \epsilon 0^* \epsilon 1^* \epsilon 1^*) \epsilon 0^* \epsilon 0^* \epsilon 1^*$ $\rightarrow \epsilon, 1, 11, 111$

$(00 \cup 01 \cup \epsilon), 001$

$(0+1)^* \rightarrow \{\epsilon, 0, 1, 00, 11, \dots\}$

$00001, 011001, 01001$

Ör: $\Sigma = \{0, 1\}$ Σ^* 01 ile başlamak zorunda

$\epsilon, 0, 1, 00, 11, 001, \dots$

Budilin regüler ifadesini yazacağız.

$(01) \cdot (0+1)^*$ $\rightarrow \{01, 010, 011, 01000\}$ olmak zorunda

Ör: 01 ile bitecek regüler ifade $(0+1)^* \cdot (01)$ $\epsilon 001, 010101, \dots$

01 alt katarın içeren dilin regüler ifadesi

$(0+1)^* (01) (0+1)^* \rightarrow \{01, 101, 001, 00101, 0011, 011, \dots\}$

$(0+1)^+ \rightarrow \{0, 1, 00, 11, 01, 10, \dots\}$

regüler ifadelerde kullanılabiliyor eğer ϵ ifade yok ise

Ör: $L = \{X W X^n\} \mid X \in (0+1)^*, W \in (0, 1)^*\}$ **quiz sorusu**

regüler ifade değil **nedeni** bellek gereksimi olduğu için regüler ifadeyi yazılmaz.

Ör: $L = \{a^n b^n\} \rightarrow \{a, ab, aabb, aaabbb, \dots\}$

regüler ifade değil çünkü bellek gereksimi olduğu için regüler ifadeyi yazılmaz.

Not $a^* b^* \neq a^n b^n$

Ör: $L = \{X^{2n+1} \mid n=0\}$ bu dilli tanıyan regüler ifadeyi yazınız

$\epsilon X, XXX, XXXXX, \dots$

$X(XX)^* \iff (XX)^* X$

Ör: $\Sigma = \{a, b, c\}$ "a" yada "c" ile başlayan kısıtlamasız sayıda "b" ile devam eden dilin regüler ifadesini yazınız.

$(a+c) b^*$

ör: $\Sigma = \{a, b\}$ en az 2 tane "a" içeren dil regüler ifade si yazınız

$$(a+b)^* a (a+b)^* a (a+b)^*$$

Tam olarak iki tane "a" içeren dilln regüler ifadesini yazınız

$$b^* a b^* a b^*$$

ör: $\Sigma = \{a, b\}$ en az bir "a" en az bir "b" içeren dilln regüler ifadesini yazınız

$$(a+b)^* a (a+b)^* b (a+b)^* + (a+b)^* b (a+b)^* a (a+b)^*$$

ör: $\Sigma = \{0, 1\}$ çift uzunluluk katarlar tanıyan dilln regüler ifade yazınız

$$(0+1)(0+1)^*$$

ör: $\Sigma = \{0, 1\}$ 6 uzunluklu katar tanıyan dilln regüler ifade yazınız

$$(0+1)(0+1)(0+1)(0+1)(0+1)(0+1)$$

en fazla 6 uzunluklu katarlar

$$(0+1+\infty)(0+1+\infty)(0+1+\infty)(0+1+\infty)(0+1+\infty)(0+1+\infty)$$

ör: (harf, rakam, ...) Harf başlayacak uzunluk kısıt yok.

$$\text{Letter}(\text{harf}) = \{a, b, c, \dots, z, A, B, \dots, Z\}$$

$$\text{Digit}(\text{rakam}) = \{0, 1, 2, \dots, 9\}$$

$$C = \{-\}$$

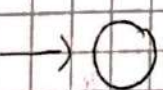
$$\text{Letter}(\text{Letter} + \text{Digit} + C)^*$$

DFA (Deterministik Finite Automata)

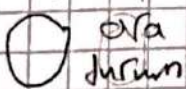
$$M_{\text{DFA}} = \langle Q, \Sigma, F, q_0, \delta \rangle$$

Q : Durumlar / Hâle
 Σ : alfabe
 F : kabul / durum kumesi
 q_0 : başlangıç / durumlar

$$\delta: Q \times \Sigma \rightarrow Q$$



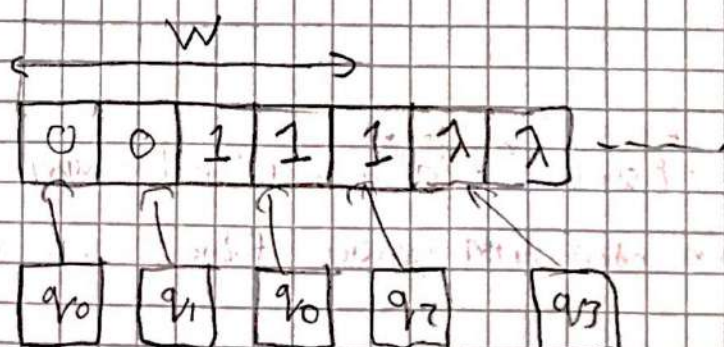
q_0 başlangıç durum



ola durum

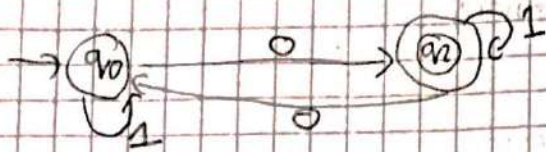


kabul durum



DFA (Deterministik Finite Automata)

(Deterministik Finite Automata)



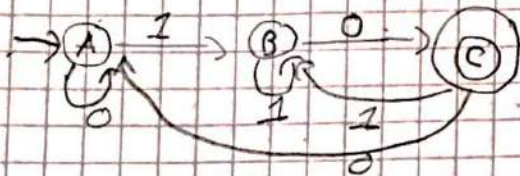
$\{0, 00, 11, 1000\}$
 tanımlı DFA tanımlı DFA tanımlı
 nemez nemez

$\delta(q_0, 1) = q_0$, $\delta(q_0, 0) = q_1$

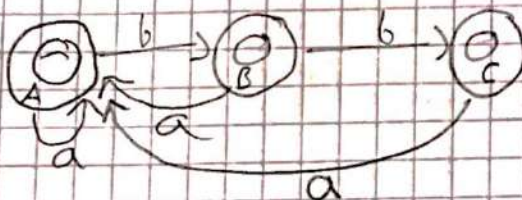
$\delta(q_1, 0) = q_0$, $\delta(q_1, 1) = q_1$

Q	0	1
A		
B		

ör $L = (0, 1)^* \{0, 1\} (0, 1)^* (0, 1)$

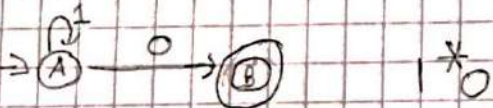


ör: $L = \{W \mid W \in \{a, b\}^*\}$; W ard arda 3 b'yi red eder



a, b ölü durum
 da kabul etmeyecek bir şekil

$- q \in Q, \delta^*(q, \epsilon) = q$



$1100 = 11 \wedge 00 = 1100 \wedge = \wedge 1100$

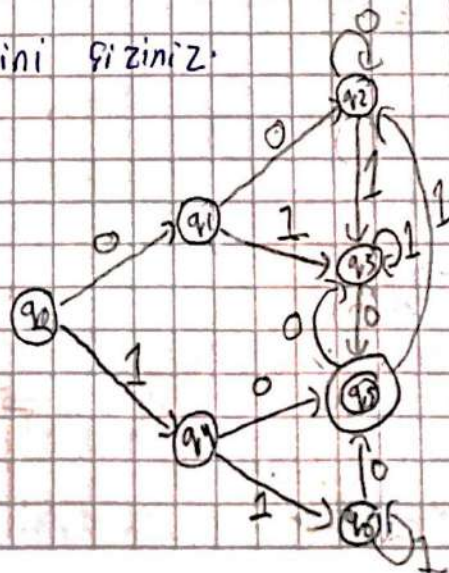
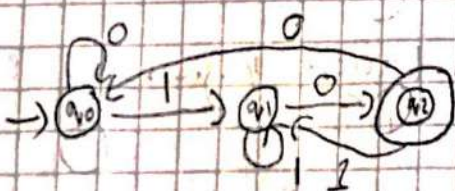
$- q \in Q, y \in \Sigma^*, a \in \Sigma$

$\Rightarrow \delta^*(q, ya) = \delta(\delta^*(q, y), a)$

$- \delta^*(q, ya) \cap F \neq \emptyset$

ör: $L = \{W \mid W, "10" \text{ ile bitiyor}\}$ DFA'sini çiziniz.

$L_{Regüler} = (0+1)^* 10$

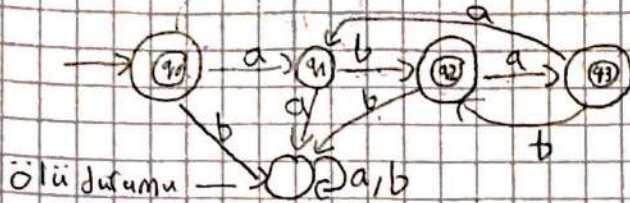


DFA (Deterministik Finite Automata)

ör: $L = (ab + aba)^*$ DFA'yi Tasarlayınız.

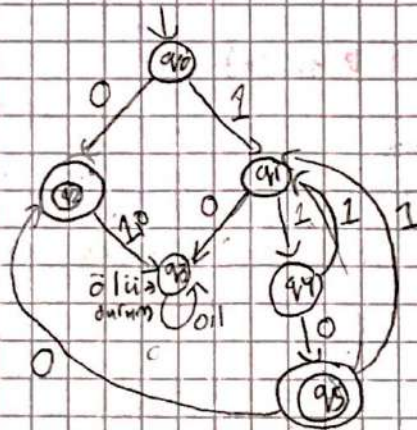
Örnek Katarları: $L = \{ \epsilon, ab, aba, abab, abaaba, ababab, \dots \}$

* "a" ile başlar * "a" yi "b" takip (q0)

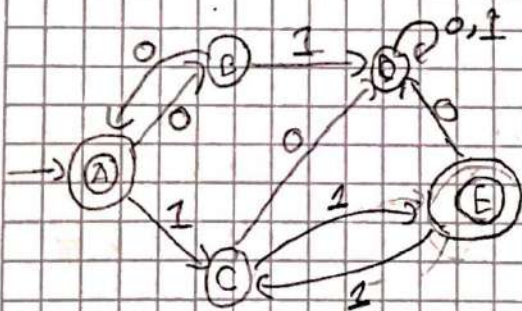


ör: $(11 + 110)^* 0$ DFA'yi tasarlayınız.

$L = \{ \epsilon 0, 110, 1100, 111100, 110110110, \dots \}$



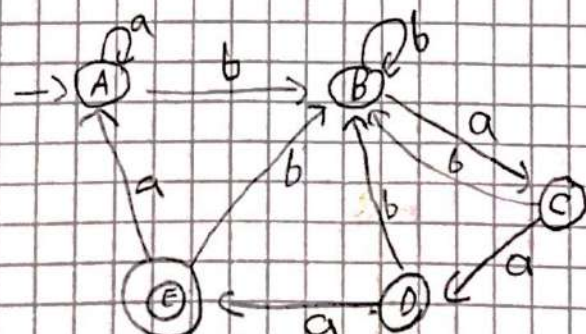
ör:



$L = \{ \epsilon, 00, 11, 0011, 1111, \dots \}$

$\Rightarrow$ Regüler ifadesi = $(00)^*(11)^*$

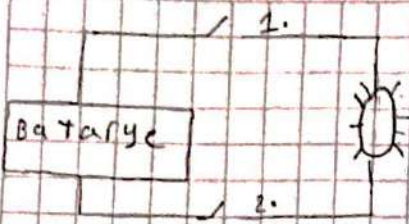
ör:



$L = \{ abbaaa, baaa, bbbbaaa, \dots \}$

$\Rightarrow$ regüler ifadesi = $(a+ib)^* \cdot baaa$

başlangıçta ϵ olmadığı için a+b kullanıdık.
Hocca sor!



DFA = $\langle Q, \Sigma, q_0, \delta, F \rangle$

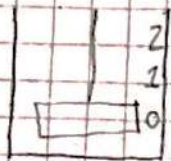
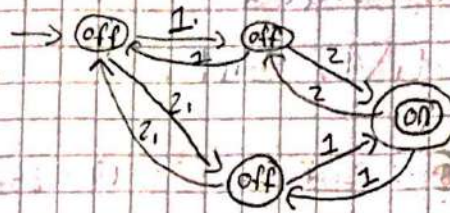
$\Sigma = \{1, 2\}$

1. anahtarın yön

2. anahtarın değiştirilmesi

$F = \{ışığın yanması\} (q_3)$

$Q = \{q_0, q_1, q_2, q_3\}$



$\Sigma = \{0, 1\}$

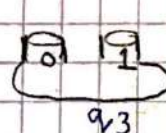
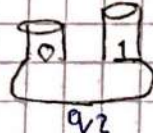
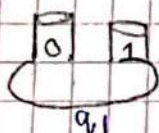
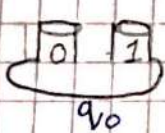
0. sürgüyü aşağı çek

1. yukarıya itmek



$F = \{elektrik gelmesi\} = \{q_1, q_2\}$

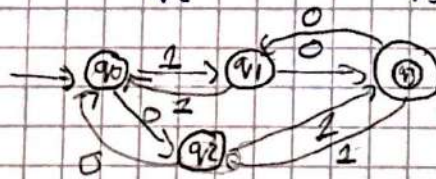
$Q = \{q_0, q_1, q_2\}$



iki buton

basılmış olması

$F = \{q_3\} = \{0, 1\}$



teorem: $U, \cap, \setminus$ kümesinin DFA'sı

Önemli

$M_1 = \langle Q_1, \Sigma, q_1, \delta_1, F_1 \rangle \rightarrow L_1$ dilin

Diğer sonuç

$M_2 = \langle Q_2, \Sigma, q_2, \delta_2, F_2 \rangle \rightarrow L_2$ dilin

$M = \langle Q, \Sigma, q, \delta, F \rangle$

$Q = Q_1 \times Q_2$ Σ aynı $q_0 = (q_1, q_2)$

$\delta((p, q), a) = (\delta_1(p, a), \delta_2(q, a))$

$L_1 \cup L_2 \Rightarrow F = F_1 \times Q_2 + F_2 \times Q_1$

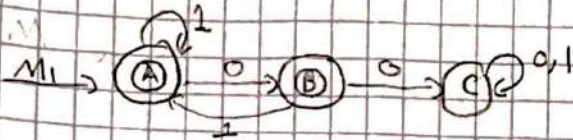
$L_1 \cap L_2 \Rightarrow F = F_1 \times F_2$

$L_1 \setminus L_2 \Rightarrow F = F_1 \times Q_2 - (F_2 \times Q_1)$

ör $L_1 = \{W \mid W, 00 \text{ alt katarların içilmez}\}$

$L_2 = \{W \mid W, 01 \text{ ile biten}\}$

$L_1 \cup L_2$, $L_1 \cap L_2$, $L_1 \setminus L_2$ bul.

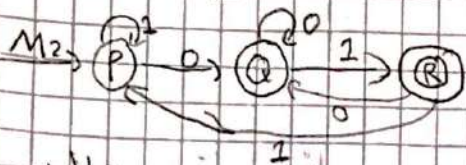


Durum
adları:

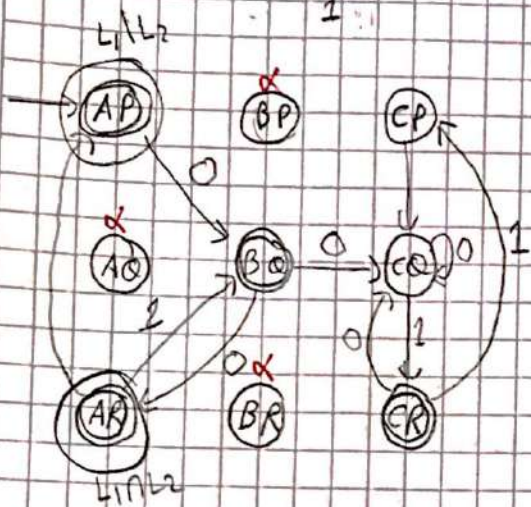
$Q_1 = \{A, B, C\}$, $F_1 = \{C\}$, $q_1 = A$

Kabul
durumu

Başlangıç durumu



$Q_2 = \{P, Q, R\}$, $F_2 = \{R\}$, $q_2 = P$



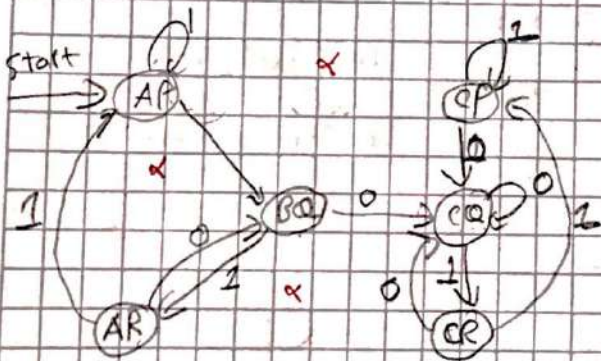
$L_1 \cup L_2$ için $F = F_1 \times Q_2 \cup Q_1 \times F_2$

$L_1 \cup L_2 = \{(A, B) \times (P, Q, R)\} \cup \{(C, A, B, C) \times (R)\}$ kabul durumları

$L_1 \cap L_2 \Rightarrow \{AR\}$

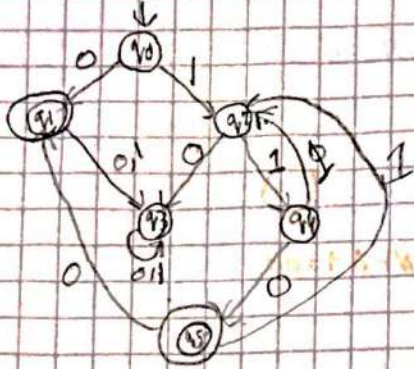
AB

$L_1 \setminus L_2 \Rightarrow \{AP\}$



Non-deterministik finite automata (NFA)

$(11+110)^*0$

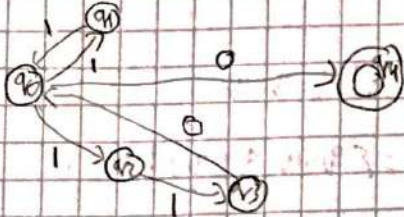


* Alfa bade bilinen her simge kullanılmak zorunda

* Her bir simge için tek çıkış olma zorunluluğu

deterministik

NFA



$W = 11011110$ DFA

NFA biçimsel tanım:

$$NFA = \langle Q, \Sigma, q_0, \delta, F \rangle$$

$$Q \times \Sigma^* \rightarrow 2^Q \quad \delta: Q \times \Sigma \rightarrow 2^Q \quad Q \in A, B \quad 2^Q = \{A\}, \{\}, \{B\}, \{A, B\}$$

$$DFA \Rightarrow Q \times \Sigma \rightarrow Q$$

$$a) \forall q \in Q \quad \delta^*(q, \epsilon) = \{q\}$$

$$b) \forall q \in Q \text{ için } y \in \Sigma^* \quad a \in \Sigma$$

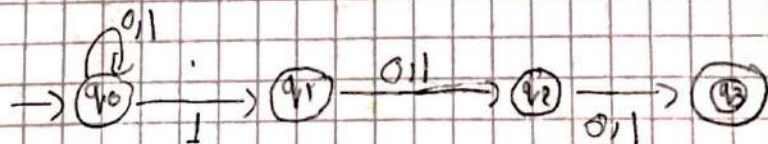
$$\delta^*(q, ya) = \bigcup_{r \in \delta^*(q, y)} \delta(r, a)$$

$$c) W = x \quad \delta^*(q_0, x) \cap F \neq \emptyset \quad W \text{ NFA } x \text{ katarı kabul edilir}$$

Örnek

q	$\delta(q, 0)$	$\delta(q, 1)$
q_0	$\{q_0\}$	$\{q_0, q_1\}$
q_1	$\{q_2\}$	$\{q_2\}$
q_2	$\{q_3\}$	$\{q_3\}$
q_3	$\emptyset$	$\emptyset$

$W = 11$ NFA



$$\delta^*(q_0, 11) = \bigcup_{r \in \delta^*(q_0, 1)} \delta(r, 1) = \delta(q_0, 1) \cup \delta(q_1, 1) = \{q_0, q_1\} \cup \{q_2\} = \{q_0, q_1, q_2\}$$

W, NFA için değil

W=111

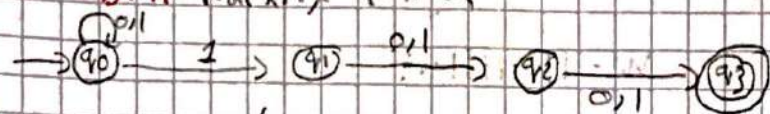
$$\delta^*(q_0, \frac{1}{y} \frac{1}{a}) = \bigcup \delta(r, 1) = \delta(q_0, 1) \cup \delta(q_1, 1) \cup \delta(q_2, 1)$$

$$\delta(q_0, 1) = \{q_0, q_1\} \cup \{q_2\} \cup \{q_3\}$$

$$\{q_0, q_1, q_2, q_3\}$$

W=111 NFA tanır

NFA → DFA karışığı çizilmez,



$$\delta(q_0, 0) = \{q_0\} \checkmark$$

$$\delta(q_0, 1) = \{q_0, q_1\} \checkmark$$

$$\delta(q_0, 0) \cup \{q_1, 0\}$$

$$\delta(q_0, q_1, 0) = \{q_0, q_2\} \checkmark$$

$$\delta(q_0, q_1, 1) = \{q_0, q_1, q_2\}$$

$$\delta(q_0, q_2, 0) = \{q_0, q_3\} \checkmark$$

$$\delta(q_0, q_2, 1) = \{q_0, q_1, q_3\}$$

$$\delta(q_0, q_1, q_2, 0) = \{q_0, q_2, q_3\}$$

$$\delta(q_0, q_1, q_2, 1) = \{q_0, q_1, q_2, q_3\}$$

$$\delta(q_0, q_3, 0) = \{q_0\}$$

$$\delta(q_0, q_3, 1) = \{q_0, q_1\}$$

$$\delta(q_0, q_1, q_3, 0) = \{q_0, q_2\}$$

$$\delta(q_0, q_1, q_3, 1) = \{q_0, q_1, q_2\}$$

$$\delta^*(q_0, q_1, q_3, 0) = \{q_0, q_3\}$$

$$\delta(q_0, q_1, q_3, 1) = \{q_0, q_1, q_3\}$$

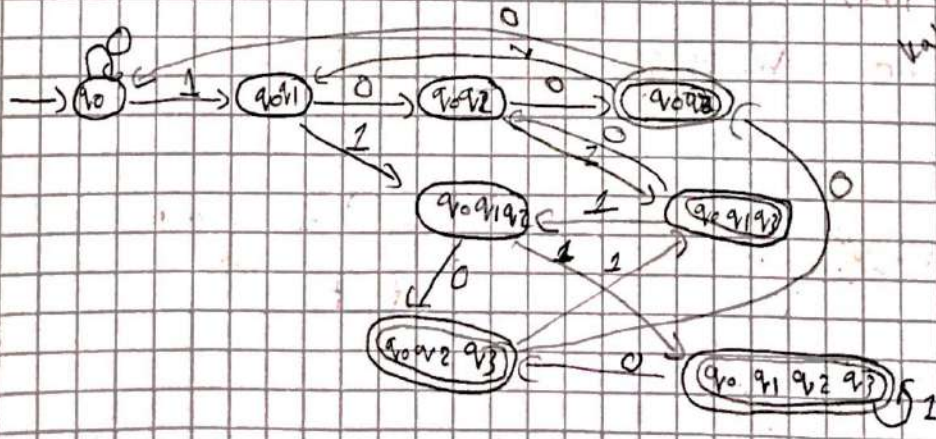
$$\delta(q_0, q_2, q_3, 0) = \{q_0, q_2, q_3\}$$

$$\delta(q_0, q_2, q_3, 1) = \{q_0, q_2, q_3\}$$

Burada işlemler yeni durumlar

çıkıyor, zaman incelemeye devam ediyor.

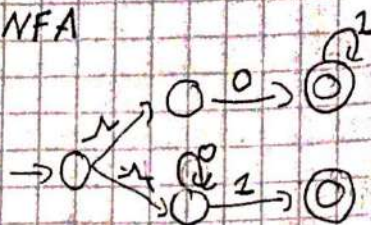
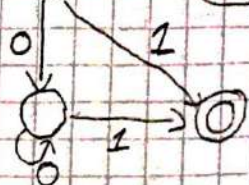
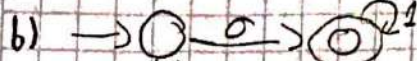
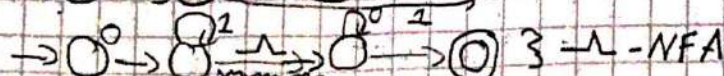
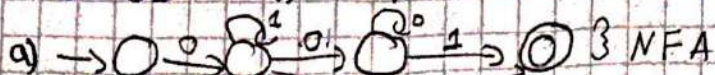
örneğin en son kabul durumu var



NFA

a) 01^*01

b) $01^* + 0^*1$



NFA

$M = \{Q, \Sigma, \delta, F, q_0\}$

* $\delta^*: Q(\Sigma \cup \{\epsilon\}) \rightarrow 2^Q$

* $q \in Q: \delta^*(q, \epsilon) = \{q\}$ $\rightarrow q$ durumunda boşluk kapanması
 $\{q \text{ durumundayken } \epsilon \text{ ile gidebileceği durumlar}\}$

* $q \in Q, y \in \Sigma^*, a \in \Sigma, \delta^*(q, ya) = \bigcup_{r \in \delta^*(q, y)} \delta(r, a)$

* $\delta^*(q_0, x) \cap F \neq \emptyset$ x kabul edilir

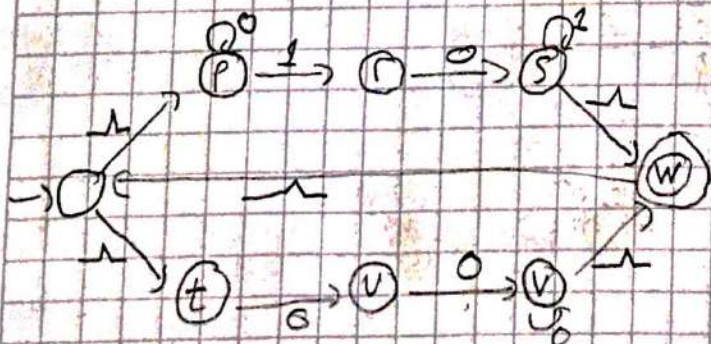
ϵ (S) in algoritme adımları

* $S, \epsilon(S)$ 'e aittir

* S 'den ϵ ile ulaştırılabilir durumlar

* Yeni durumlarda ϵ ile ulaştırılabilir durumlar

* $\epsilon(S)$ elemanları değişmeyecek kadar adımları tekrarla.



$$\delta^*(q_0, \epsilon) = \{q_0\}$$

$$\{q_0, p, t\}$$

$$\delta^*(q_0, 0) = \{q_0, p, t\}$$

$$\bigcup_{r \in \delta^*(q_0, \epsilon)} \delta(r, 0) = \{q_0, p, t\}$$

devamı!

$$\neg (S(q_0, 0) \cup S(p, 0) \cup S(t, 0)) = \neg (p, u) = (p, u)$$

$$\emptyset \cup \{p\} \cup \{u\} \quad \{p, u\}$$

or $0^*(01)^*0^*$



* q	S(q, 0)	S(q, 1)	S*(q, 0)	S*(q, 1)
A	{B}	{A}	{A, B, C, D}	{}
B	{D}	{C}	{C, D}	{}
C	{}	{B}	{B}	{C, D}
D	{}	{D}	{}	{}

$$S^*(A, 0) = \neg \left(\bigcup S(p, 0) \right)$$

$$\neg (S^*(A, 1))$$

$$\neg (A)$$

$$S^*(A, 1) = \neg (V S(p, 1))$$

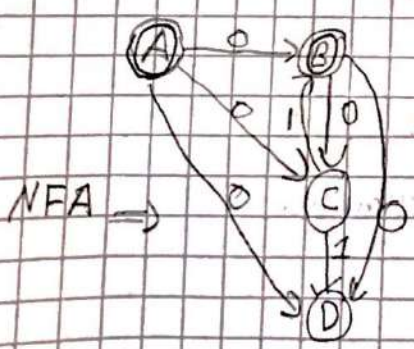
$$\neg (S^*(A, 1))$$

$$= \neg (\emptyset \cup \emptyset \cup \emptyset) = \neg (\emptyset) = \emptyset$$

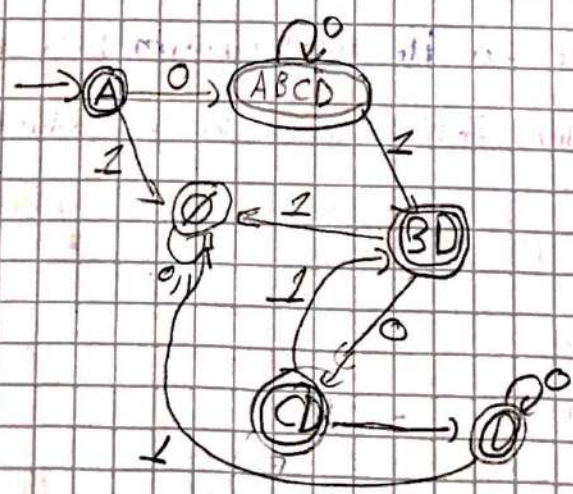
{A, B, D}

$$\neg (S(A, 0) \cup S(B, 0) \cup S(D, 0))$$

$$\neg (A, C, D) = \{A, C, D, B\}$$

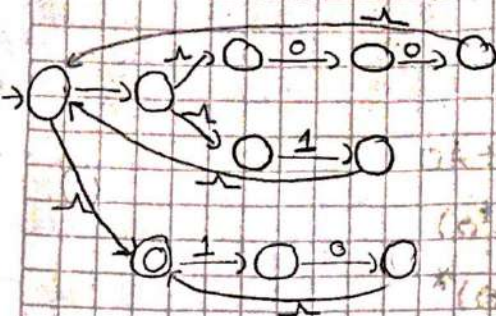


DFA =>

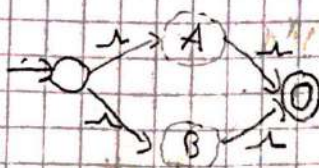


Λ - NFA

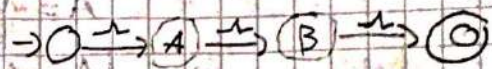
$$r_i = \underbrace{(00+1)}_A^* \underbrace{(10)}_B^*$$



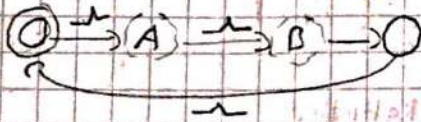
* A+B istendiğinde



* A.B istendiğinde



* (AB)* istendiğinde

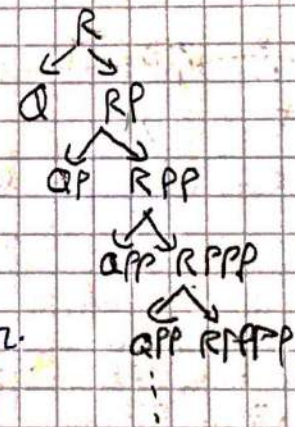


P, Q, R aynı alfabe de tanımlı, regüler ifade olsun

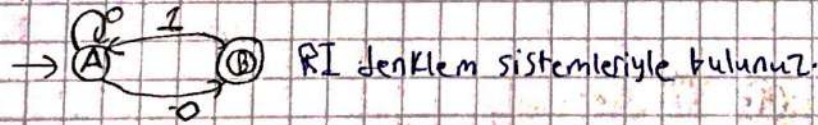
P, Λ içermiyorsa

R = Q + RP biçiminde yazabiliyorsa

R = Q.p* olarak ifade edebilir.



ör:



RI denklem sistemleriyle bulunuz.

$$A = \Lambda + A0 + B1 \quad (1)$$

$$B = A0 \quad (2) \quad T(M) = B$$

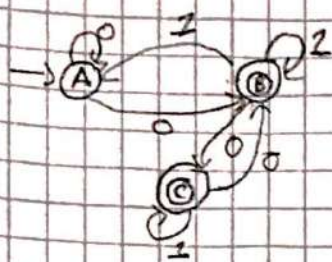
$$A = \Lambda + A0 + A01$$

$$A = \Lambda + A(0+01)$$

$$A = \Lambda(0+01)^*$$

$$B = \Lambda(0+01)^*0$$

or



$$A = \lambda + A_0 + B_1 \quad (I)$$

$$B = B_1 + C_0 + A_0 \quad (2)$$

$$C = C_1 + \beta_0 \quad (3)$$

$$T(M) = B + e$$

$$C = B_0 + C_1$$

$$R \quad R + R$$

$$C = B \cap I^*$$

$$B = B_1 + B_0 I^* O + A O$$

$$B = A_0 + B(1 + 0.1^*0)$$

$$B = A_0(I + 0.1^* 6)^*$$

$$A = -L + A_0 + A_0(I + 0.1^* 0)^* 1$$

$$A = \underline{1} + A(O + O(1 \oplus 01^*0))^*1$$

$$A = -\lambda(0 + 0(1 + 0 \cdot 1^* 0) \cdot 1)^*$$

$$B = (0 + 0(1 + 01^*0)^*1)^*0(1 + 01^*0)^*$$

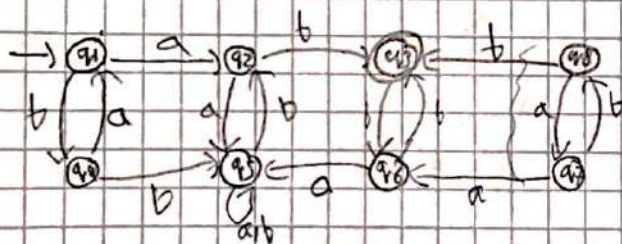
$$c = (0 + 0(1 + 01^*0)^*1)^*0(1 + 01^*0)^*01^*$$

1) + cevap

Durum Sayısı indirgeme

- 2) Başlangıç durumlar itibaren ulaşılamaya durumlar kabılır.

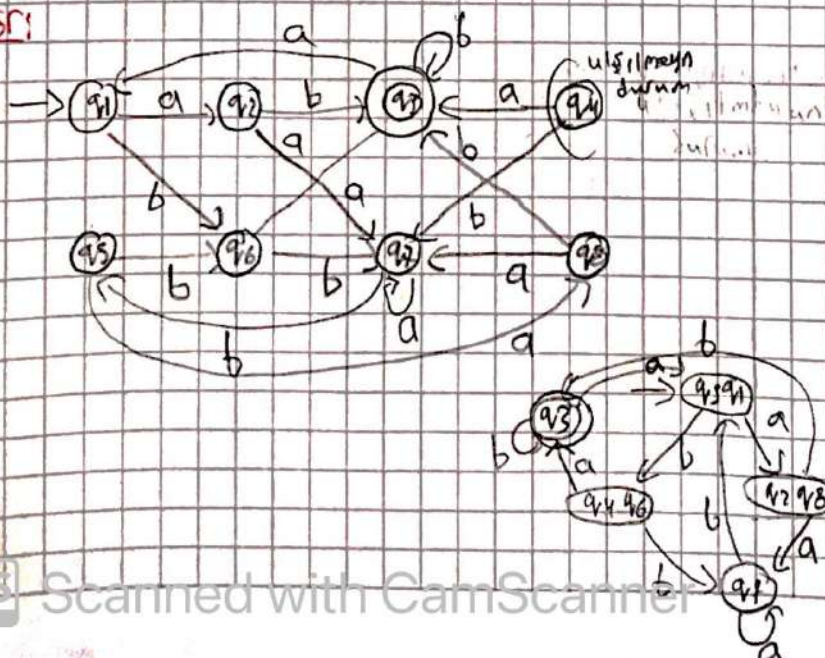
- 2) Derik durumlar birleştirilir.



KB, 97 414511 mag or.

q2 ve q6 denklemleri ile

ॐ



92	X								
93	X	X							
94	X	X	X						
95	/	X	X	X					
96	X	X	X	/	X				
97	X	X	X	X	X	X			
98	X	/	X	X	X	X	X		
	91	92	93	94	95	96	97	98	

Grammeler

$$G = \langle V_N, V_T, P, S \rangle$$

non
Terminaller

Terminaller

türetimi
kısıtlı lara

başlangıç VT

$$S \in V_N$$

$$L(G) = \{ W \mid W \in V_T^*; S \Rightarrow^* W \}$$

non-terminal olmaz

$$V = V_N \cup V_T$$

$$P: \alpha \Rightarrow \beta$$

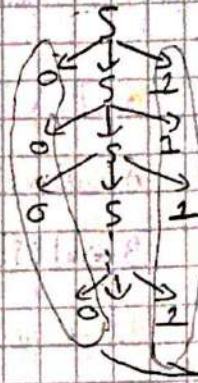
$$\alpha \in V^+, \beta \in V^*$$

$$\text{Ör: } G = \langle V_N, V_T, P, S \rangle$$

$$V_N = \{ S \}, V_T = \{ 0, 1 \}$$

$$P: \begin{cases} S \Rightarrow 0S1 \\ S \Rightarrow 01 \end{cases} \Rightarrow \begin{matrix} 0S1 \\ 00S11 \\ 000S111 \end{matrix} \Rightarrow W$$

$W = 000111$



türetim ağacı

tümcesel form

tümce

Tür-0 grammerler

$$\alpha \Rightarrow \beta, \alpha \in V^+, \beta \in V^* \text{ ise tür-0 grammerlerdir.}$$

* Güç fazla

* Tasarım zor

* Rekürsif Sayılabilir diller

* Turing makinaları

Tür-1 grammerler

$$\alpha \Rightarrow \beta, \alpha \in V^+, \beta \in V^*, |\alpha| \leq |\beta|$$

* Bağlamdan bağımlı diller (non-context-free grammer)

* LBA

Tür-2 grammerler

$$\alpha \Rightarrow \beta, A \Rightarrow \beta, A \in V_N, \beta \in V^*$$

* Context-free grammer

* Push-down automata (PDA)

Tür-3 gramer (ler) (reguler diller - reguler gramer)

$A \Rightarrow aB$, $A \Rightarrow a$, $A \Rightarrow \Lambda$, $A, B \in V_N$, $a \in V_T$

Sağdan türetilen

Soldan türetilen

DFA, NFA, L-NFA

$A \Rightarrow WB$

$A \Rightarrow BW$

$A \Rightarrow W$

$A \Rightarrow W$

$W \in V_T$

$A \Rightarrow \Lambda$

$A \Rightarrow \Lambda$

Ör: $S \Rightarrow LAaR$

$Aa \Rightarrow aaA$

Ör: $P: S \Rightarrow aB \mid bA$

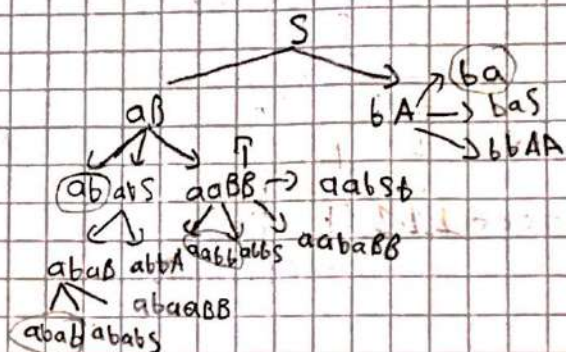
$A \Rightarrow a \mid aS \mid bAA$

$B \Rightarrow b \mid bS \mid aBB$

tür-2 gramer dir

Esit sayıda

"a" ve "b" içeren diller tanımlı



Tür-0 $AR \Rightarrow BR$

gramer dir $aB \Rightarrow Ba$

$LB \Rightarrow LA$

$aC \Rightarrow Ca$

$LC \Rightarrow \Lambda$

Ör: $P: S \Rightarrow OS \mid BA \mid \Lambda$

$A \Rightarrow OB$, $B \Rightarrow IS$

tür-3 gramer dir sağ taraflı.

$G \langle V_N, V_T, P, S \rangle$

$M = \langle Q, q_0, \Sigma, \delta, F \rangle$

$V_N = \{S, A, B\}$

$\Sigma = V_T$

$V_T = \{0, 1\}$

$Q = V_N \cup \{c\}$

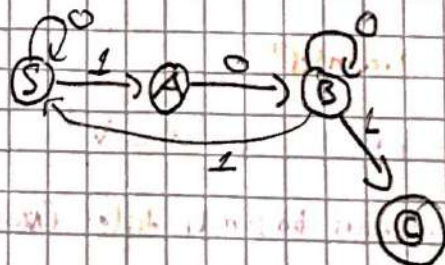
$S \Rightarrow OS \mid IA$

$F = \{c\}$

$A \Rightarrow OB$

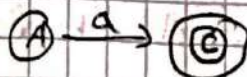
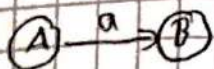
$q_0 = \{S\}$

$B \Rightarrow OB \mid IS \mid I$

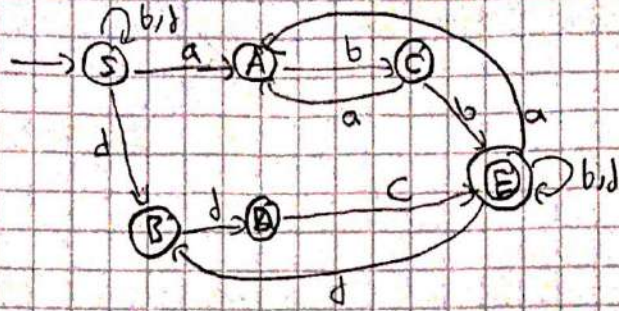


Si $\forall A \Rightarrow aB$ kural için

Si $\forall A \Rightarrow a$ için



ör1



$$S \Rightarrow bS \mid dS \mid aA \mid dB$$

$$A \Rightarrow bc$$

$$B \Rightarrow dD$$

$$C \Rightarrow aA \mid bE$$

$$D \Rightarrow cE$$

$$E \Rightarrow bE \mid dE \mid b \mid d \mid dB \mid aA$$

ör: $L = \{a^n b^n \mid n \geq 0\}$ gramerini yazı

$$S \Rightarrow \epsilon \mid lab \mid aSb$$

ör: $L = \{W \mid W \in \{0,1\}^*, n_0(W) = n_1(W)\}$

$$S \Rightarrow \epsilon \mid 0S1 \mid 1S0 \mid 01S \mid 10S \mid SS$$

ör: $L = \{WW^r \mid W \in \{a,b\}^*, W^r: \text{reverse of } W\}$ gramerini tasarlayınız.

$$\begin{matrix} W_r = 01 \\ W = 10 \end{matrix} \left\{ \begin{matrix} 0110 \\ 1001 \end{matrix} \right. \quad S \Rightarrow 0S0 \mid 1S1 \mid \epsilon$$

$$\begin{matrix} W_r = 10 \\ W = 01 \end{matrix} \left\{ \begin{matrix} 1001 \\ 0110 \end{matrix} \right.$$

$$\begin{matrix} W_r = 11 \\ W = 11 \end{matrix} \left\{ \begin{matrix} 11011 \\ 11101 \end{matrix} \right.$$

Aritmetik ifadeler için gramerler

$$S \Rightarrow S+S \mid S-S \mid S * S \mid S / S \mid (S) \mid a$$

Belirsiz gramerdir bir ifade elimiz.

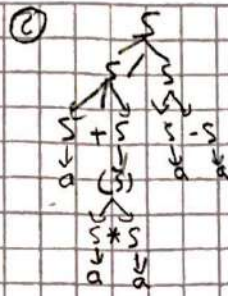
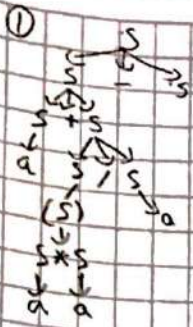
$W = a + (a * a) / a - a$ W-Karakter katar gramer ait mi?

$$① S \Rightarrow S-S \Rightarrow S+S-S \Rightarrow a+S-S \Rightarrow a+S/S-S \Rightarrow a+(S)/S-S \Rightarrow$$

$$\Rightarrow a+(S * S)/S-S \Rightarrow a+(a * S)/S-S \Rightarrow a+(a * a)/a-a$$

$$② S \Rightarrow S/S \Rightarrow S+S/S \Rightarrow a+S/S \Rightarrow a+(S)/S \Rightarrow a+(S * S)/S \Rightarrow$$

$$\Rightarrow a+(a * S)/S \Rightarrow a+(a * a)/S \Rightarrow a+(a * a)/S-S \Rightarrow a+(a * a)/a-a$$



$$S \Rightarrow S+T \mid S-T \mid T$$

$$T \Rightarrow T * F \mid T / F \mid F$$

$$F \Rightarrow (S) \mid a$$

Theorem:

$\forall L_1, L_2 \text{ CFL is } L_1 \cup L_2, L_1 \cap L_2, L_1^* \text{ CFL}$

$$L_1 \Rightarrow G_1 = \langle V_1, \Sigma, S_1, P_1 \rangle$$

$$L_2 \Rightarrow G_2 = \langle V_2, \Sigma, S_2, P_2 \rangle$$

a) $L_1 \cup L_2$ is in;

$$G_u = \langle V_u, \Sigma, S_u, P_u \rangle$$

$$V_u = V_1 \cup V_2 \cup \{S_u\} \rightarrow \text{ortak}$$

$$P_u = P_1 \cup P_2 \cup \{S_u \Rightarrow S_1 \mid S_2\}$$

b) $G_c = \langle V_c, \Sigma, S_c, P_c \rangle$

$$V_c = V_1 \cup V_2 \cup \{S_c\}$$

$$P_c = P_1 \cup P_2 \cup \{S_c \Rightarrow S_1 \cdot S_2\}$$

c) $G^* = \langle V, \Sigma, S, P \rangle$

$$V = V_1 \cup \{S\}$$

$$P = P_1 \cup \{S \Rightarrow S_1 S_1^{-1}\}$$

Ör: $L = \underbrace{(011+1)}_B^A \underbrace{(01)}_C^* \text{ CFG yazınız.}$

$A \Rightarrow 0111$

$D \Rightarrow 01$

$B \Rightarrow AB | \perp$

$C \Rightarrow DC | \perp$

$S \Rightarrow B, C$

$B \Rightarrow AB | \perp$

$C \Rightarrow DC | \perp$

$A \Rightarrow 0111$

$D \Rightarrow 01$

Ör: $L = \{0^i 1^j 0^k \mid j \geq i + k\}$ CFG ile tanımlı

$\underbrace{0^i}_{L_1} \underbrace{1^j}_{L_3} \underbrace{0^k}_{L_2} \quad S \Rightarrow ABC$

$L_1 = \{0^i \mid i \geq 0\} \Rightarrow A \Rightarrow 0A1 | \perp$

$L_2 = \{0^k \mid k \geq 0\} \Rightarrow C \Rightarrow 1C0 | \perp$

$L_3 = \{1^m \mid m \geq 1\} \Rightarrow B \Rightarrow 1B | 1$

Ör: $G = \langle V_N, V_T, S, P \rangle$

$L = \{a^i b^j c^k \mid i, j, k \geq 1\}$

$V_N = \{S, A, B\}$

$V_T = \{a, b, c, d\}$

$P: S \Rightarrow AA \mid d$

$A \Rightarrow aAd \mid A \mid bBc$

$B \Rightarrow bBc \mid Bc \mid \perp$

Ör: $V_N = \{S, x, y, z\} \quad V_T = \{a, b, c\}$

$P, S \Rightarrow xy$

$L = \{a^i b^j c^k \mid i, j, k \geq 1\}$

$x \Rightarrow axbb \mid azbb \mid ab$

$y \Rightarrow cy \mid c$

$z \Rightarrow zb \mid xb$

Gramellerin Sadeleştirilmesi

- 1 Doğrudan rekursiflik yok et $C \Rightarrow SA \mid S \Rightarrow AS$
- 2 Yok edilebilir değişkenler kaldırılır. (ϵ ; başlangıç NT olmalıdır.)
- 3 Birim türetme kurallarını kaldır. $(A \Rightarrow B^C)$
 $B \Rightarrow \epsilon$
- 4 Yalansız değişken/terminalleri kaldır.

Yok edilebilir değişken

* $A \Rightarrow \epsilon$ ise yok edilebilir. Cigerisinde A olan kurallar tekrar düzenlenir

$$00. S \Rightarrow ACA$$

$$S \Rightarrow ACA \mid CA \mid AA \mid AC \mid A \mid C \mid \epsilon$$

$$A \Rightarrow B \mid C \mid aAa$$

$$A \Rightarrow B \mid C \mid aAa \mid aa$$

$$T_u = \{C, A, S\}$$

$$B \Rightarrow bB \mid b$$

$$B \Rightarrow bB \mid b$$

$$C \Rightarrow cC \mid \epsilon$$

$$C \Rightarrow cC \mid c$$

Kaldırma Kuralları

$$S \Rightarrow ACA \mid CA \mid AA \mid AC \mid aAa \mid aa \mid bB \mid b \mid cC \mid c \mid \epsilon$$

$$A \Rightarrow aAa \mid aa \mid \underbrace{bB \mid b}_B \mid \underbrace{cC \mid c}_C$$

$$B \Rightarrow bB \mid b$$

$$C \Rightarrow cC \mid c$$

Yalansız değişken/terminal

C, D, terminal üretilir

$$S \Rightarrow B \mid A \mid B \mid c$$

$$T_u = \{B, S, A, E, F\}$$

$$A \Rightarrow aA \mid aF$$

$$B \Rightarrow c \mid b$$

$$E \Rightarrow c \mid d$$

$$F \Rightarrow c \mid a \mid b \mid d$$

$$E \Rightarrow aA \mid bSA$$

$$F \Rightarrow bB \mid d$$

$$S \Rightarrow B \mid B \mid S$$

$$B \Rightarrow b$$

en sade hali